

AD 502 – Machine Learning

Pre-Reading Material – Lecture 26

Gaussian Mixture Algorithm

Introduction

A **Gaussian Mixture Model (GMM)** is an **unsupervised machine learning technique** used for clustering data. It assumes that the dataset is generated from a combination of several **Gaussian (Normal) distributions**.

In simple words:

GMM = Multiple Gaussian distributions + Probability-based clustering

Unlike K-Means, GMM does not simply assign a data point to one cluster. Instead, it provides the **probability of a data point belonging to each cluster**.

Gaussian Distribution

A **Gaussian distribution**, also called a **Normal distribution**, is a probability distribution commonly represented by a **bell-shaped curve**.

In a GMM, each Gaussian distribution represents a possible cluster in the dataset. Different Gaussian distributions can have different centers, spreads, and orientations.

Mixture Model

A **mixture model** assumes that the dataset contains several underlying groups.

For example, a customer dataset may contain:

- Low-spending customers
- Medium-spending customers
- High-spending customers



GMM represents these groups using different Gaussian distributions and determines how strongly each data point belongs to each group.

Soft Clustering

One of the most important features of GMM is **soft clustering**.

In **hard clustering**, such as K-Means, a data point is assigned to only one cluster.

In GMM, a data point can have different membership probabilities.

For example:

Cluster 1: 70%

Cluster 2: 25%

Cluster 3: 5%

Therefore, GMM is useful when clusters **overlap** and a clear boundary between groups is difficult to establish.

How GMM Works

GMM commonly uses the **Expectation-Maximization (EM) algorithm**.

The basic process is:

1. **Initialize** the Gaussian components.
2. **Estimate** the probability that each data point belongs to each component.
3. **Update** the characteristics of the components based on these probabilities.
4. **Repeat** the process until the model becomes stable.

Thus:

Estimate membership → Update clusters → Repeat

GMM vs K-Means

K-Means

Hard clustering

Assigns each point to one cluster

Uses cluster centroids

Less suitable for overlapping clusters

Simple and fast

GMM

Soft clustering

Gives probability for each cluster

Uses Gaussian distributions

Better suited for overlapping clusters

More flexible

Applications

GMM can be used for:

- Image segmentation
- Speech recognition
- Customer segmentation
- Anomaly detection
- Pattern recognition
- Computer vision
- Medical data analysis

Key Takeaways

- GMM is an **unsupervised clustering technique**.
- It represents data using a **mixture of Gaussian distributions**.
- It performs **probability-based or soft clustering**.
- It can handle **overlapping clusters**.
- The **EM algorithm** is commonly used to train GMM.
- Important concepts include **Gaussian components, probability, mean, covariance, and cluster membership**.